GRAFT: Symmetry-Aware Atom Mapping through Continuous Fragment Growth

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Atom mapping for mechanistic exploration requires retaining distinct correspondences while controlling redundant atom permutations. We introduce GRAFT, a symmetry-aware atom-mapping algorithm based on continuous fragment growth. A fragment carries multiple compatible placements as it expands, continues growing after a placement becomes unique, and branches on distinct placements at saturation. Context-preserving deduplication merges equivalent candidates and compatible cumulative states while retaining correlated symmetry and the decisions supporting each mapping family. On all 1,851 Golden benchmark records, bidirectional GRAFT with one seed ordering per single-edge cut recovers a verified reference witness in 1,833 cases (99.03%); three orderings recover 1,836 (99.19%). One ordering uses 7.50-fold less recorded search CPU than ten on 1,807 mutually completed reactions. On 140 native XYZ/WBO endpoint pairs, GRAFT returns full explicit-atom mappings throughout and retains additional low-event alternatives relative to saved SLAP outputs. This collection has no reference annotations and therefore does not establish chemical accuracy. A supporting SLAP ablation demonstrates that sweeping single-edge constraint relaxations improves reference recovery. GRAFT provides a structured representation of mapping alternatives through continued growth, conditional branching, and deduplication, with explicit limits from finite search and verification budgets.

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1 Introduction

Atom-to-atom mapping connects a reaction's endpoint structures to an account of which atoms persist and which relationships change. The selected correspondence influences reaction-center identification, template extraction, and endpoint preparation for computational reaction studies. Curated benchmarks also show that evaluation requires careful treatment of molecular representation, incomplete records, and chemically equivalent assignments (3).

Modern approaches provide mappings through distinct routes. RXNMapper extracts correspondence information from attention in a language model trained on unannotated reactions (5). LocalMapper learns from chemist-labeled reactions through a human-in-the-loop procedure (1). SLAPMapper combines Weisfeiler–Lehman-like label refinement with sequential linear assignment problems and exposes alternative mappings (2). These approaches motivate evaluating both recovery of annotated correspondences and the alternative assignments retained by a search.

For mechanistic exploration, an alternative can be useful even when it is not the lowest-edit output. Conversely, many atom-index permutations need not represent distinct chemistry. A useful search representation must distinguish alternative structural decisions from redundant realizations and preserve correlations between atoms that move together. Ranking one representative cannot characterize the entire family it represents.

Here we introduce GRAFT, an atom-mapping algorithm that constructs fragments continuously on weighted endpoint graphs. Its central design couples three operations: extending a fragment while retaining compatible placements, branching on distinct placements when growth saturates, and deduplicating candidates and cumulative states without erasing context needed by later decisions. Completed mapping families preserve correlated atom permutations and their search histories. We first illustrate these operations, then evaluate Golden reference recovery, the cost of seed diversity, and alternatives on native coordinate and bond-order inputs. Single-edge cuts provide a bounded way to relax growth constraints; a supporting SLAP ablation tests the usefulness of this perturbation strategy in a separate mapper.

2 Methods

GRAFT alternates local fragment growth with decisions about how saturated fragments are placed (Figure 1). Ambiguity remains represented during extension, and whole-mapping branches inherit the constraints of earlier decisions. Deduplication acts within this search and on its completed output, preserving the distinctions required for valid continuation and family membership.

2.1 Weighted endpoints and mapping scope

Let V_R, V_P denote source and target atom sets, Z_R, Z_P their element labels, and W_R, W_P their symmetric bond-weight matrices. The search constructs an injective element-compatible correspondence $m : D_R \subseteq V_R \to V_P$. Equal-composition endpoints admit a full bijection; partial coverage remains explicit for incomplete records. Hydrogen atoms participate in search. Coordinates accompany native endpoints but do not determine the growth decisions considered here.

An active graph is obtained using bond floor $b = 0.2$. Extension checks use the supplied numerical weights with tolerance τ. Golden inputs use formal bond-order values from molecular records; the coordinate collection uses cached continuous Wiberg bond-order (WBO) matrices. These representations are distinct. Continuous fragment growth refers to continued incremental extension while compatible placements evolve; the method also accepts continuous weights.

2.2 Continuous fragment growth

A fragment starts at one source atom with a compressed collection of compatible target placements. Its frontier is a priority queue of active source edges, ordered by decreasing weight with deterministic tie handling. For a proposed source atom n and target image v, a placement must preserve element identity and injectivity and satisfy, for every active edge (n, u) to the current fragment F,

$$W_P[v, m(u)] \geq b, \qquad |W_R[n, u] - W_P[v, m(u)]| \leq \tau. \tag{1}$$

All such edges are checked, rather than only the queue edge initiating the attempt. Source nonedges are not local negative constraints; additional target bonds can later be classified as formed bonds.

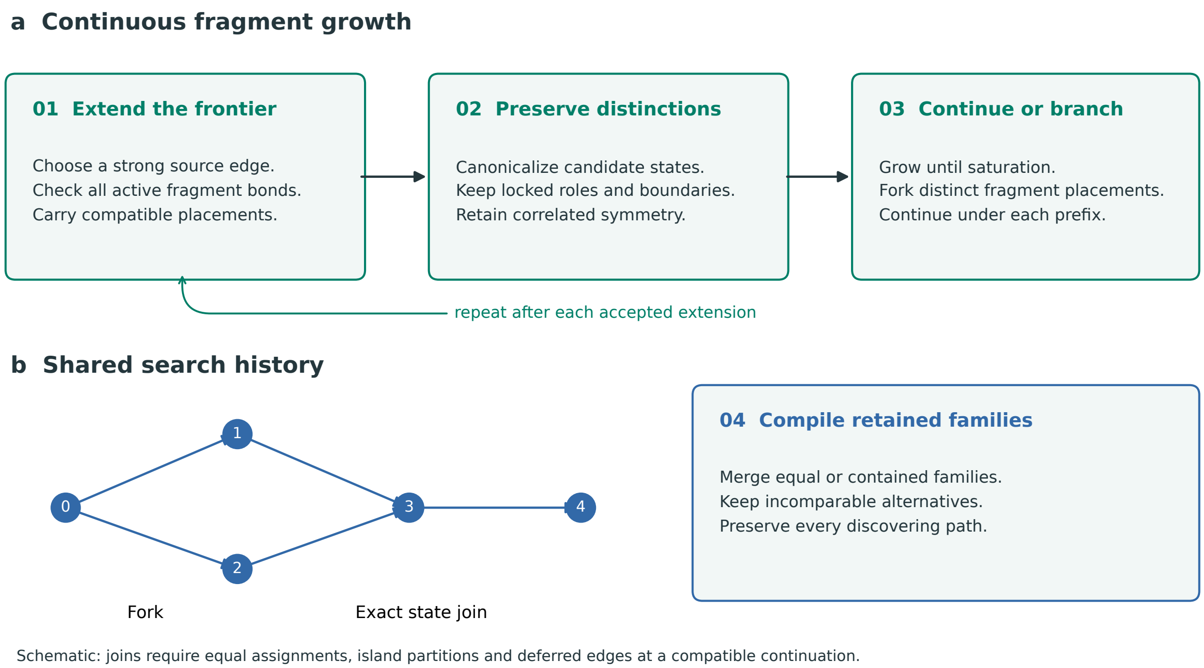

Figure 1 Continuous growth and shared search history. (a) Extension alternates with candidate canonicalization until saturation; distinct placements continue as separate mapping branches. (b) Schematic fork and reconvergence. Joins require equal cumulative states at a compatible continuation within one context. This diagram is conceptual, not a measured reaction trajectory.

A candidate stores a representative mapping, unresolved assignment blocks, and symmetry evidence. If an extension touches an unresolved block, compatibility requires a jointly consistent injective assignment. A failed arbitrary representative is insufficient to reject the compressed candidate. This local support calculation has an implementation budget and is distinct from enumerating complete mappings.

When at least one extension survives, growth commits the enlarged fragment and continues with the surviving placements. A unique placement does not end growth early. If no placement supports an extension, that edge becomes a deferred boundary and growth continues elsewhere. Reaching an already mapped atom triggers an attempt to absorb its entire locked island, verifying the added relationships and respecting existing assignments.

At saturation, distinct placements yield separate branches of the whole-mapping search. Alternatives therefore exist within a growing fragment and among sequences of saturated fragment decisions. Growth is conditional and greedy along each trajectory: accepting an extension filters candidates that cannot support it. Every possible fragment boundary is not explored, so seed orderings and cut conditions affect coverage.

2.3 Context-preserving deduplication

Deduplication operates at three levels (Figure 1). During local growth, colored graph certificates encode target structure, candidate roles, assignment pools, and locked images. Canonical labeling uses nauty (4). The configured bond-weight coloring defines the symmetry model; numerical compatibility remains subject to Equation (1). Deferred edges and the current one-hop boundary preserve distinctions relevant to later growth. Merging retains symbolic variation without computing a full automorphism group at every extension. Exact conditioned generators are finalized on recorded transitions needed by the returned results.

At the whole-mapping level, cumulative state contains the representative assignment, normalized island partition, and deferred edges. Equal states can reconverge at the same continuation step under the same cut/seed context. Both incoming histories remain connected to their shared continuation in a directed acyclic graph. Equal mappings alone do not justify discarding different fragment relations; different cut contexts are not joined solely because assignments agree.

For completed supported relations, downstream compilation represents a family by a representative m_0 and permitted target actions G:

$$\mathcal{F} = \{g \circ m_0 : g \in G\}. \tag{2}$$

Equality and containment checks merge equal or subsumed families while retaining provenance; incomparable families remain distinct. These checks require the complete relation and do not justify a more aggressive quotient of incomplete states. Atom orbits are summaries, not independent permissions to permute atoms. An earlier fragment action must transport dependent later assignments and generators together. Optional mechanism grouping and geometry-based selection consume these relations downstream.

2.4 Seed orderings, sweeps, and budgets

One benchmark seed setting denotes one ordering of source atoms, not one atom-pair assignment. The scheduler visits eligible seeds under each branch's constraints, repeating passes while progress remains possible. A sweep includes the uncut graph and each eligible single-edge deletion. Cuts change growth constraints; event evaluation uses original endpoint matrices. Deterministic cut-specific streams derive from root seed 42.

The seed ablation retains both directions, explicit H, $\tau = 1.0$, and branch cap 100 while varying only the number of orderings per cut among 1, 2, 3, and 10. Shorter lists are prefixes of the same deterministic sequence. Bidirectional output is a union of independently returned alternatives, without splicing branches.

Caps apply to live compressed candidates during growth and to post-deduplication branch admission. An overflowing parent subtree is discarded while siblings survive; exactly 100 branches are allowed. A cap can coexist with a complete mapping. The seed experiments use 300-second search watchdogs and separate 240-second analysis watchdogs. Bounded symbolic checks can remain unresolved even when analysis exits normally.

Each swept run removes at most one source edge; the union also includes the uncut run. This bounds the perturbation per trajectory without assuming that a reaction changes only one bond. The separate adaptive no-sweep experiment changes scheduler and cap policy as well as cut usage; its protocol and results are reported in the supporting information (Appendix A.5).

2.5 Golden benchmark and reference recovery

We use all 1,851 original Golden records (3), with dataset revision `793475e`. Reference labels are removed before search; records are not removed or repaired to improve compatibility. Explicit-H expansion and formal bond-order matrices provide GRAFT inputs. No coordinate optimization is performed for Golden.

Correctness is equality of the annotated heavy-atom correspondence relation, including unmatched atoms, modulo chemical automorphisms of both endpoints. Evaluator colors retain element, charge, isotope, hydrogen count, and stereochemistry, with bond-order and bond-stereochemistry relations. Returned chemistry must remain unchanged. H identity is not scored as reference provenance, although H participates in search.

Recovery requires a verified reference witness among retained alternatives. Verified absence concerns saved output. Unresolved verification establishes neither recovery nor absence. Failures and unknowns remain in the denominator. Family recovery, first-output correctness, representative top-k, and event-window recovery are distinct metrics.

2.6 Supporting single-edge ablation in SLAP

We test SLAPMapper 1.0.0 at revision `ea248fd`, retaining native alternatives (2). The new sweep uses one original ordering, both directions, and binary and weighted modes. Each variant runs the uncut input and every single source-edge deletion, rebuilding initialization after each cut. Explicit H and native element-balancing placeholders remain. Output labels are restored to the original endpoints before evaluation.

Cuts perturb SLAP's graph and objective, rather than its internal assignment or pruning rules; their effect is not assumed identical to GRAFT's fragment constraints. References never select cuts or stop searches. Each case/direction/mode variant has a 300-second watchdog across its cuts. We distinguish the uncut control, standalone sweep, and union with an earlier expanded-SLAP experiment. Default competitors appear in the supporting information without equating their cardinalities or search budgets to the sweeps.

2.7 Native XYZ/WBO collection and shared score

The second collection comprises 140 selected elementary steps with 18–137 explicit atoms per endpoint, including multicomponent and organometallic structures. Each pair has equal elemental composition. No annotated reference mappings exist. Searches consume original coordinates and cached WBOs without previous predictions or reference TS structures.

The native comparison uses GRAFT on WBO graphs and SLAP's `map_3d` interface on original component XYZ files, with native distance-based binary connectivity and bond scale 1.2. These workflows have different representations. Saved full assignments are then scored on the same WBO matrices, including H. A bond is present above 0.2; a preserved bond contributes an order-change event when its WBO difference exceeds 0.5:

$$E(m) = N_{\text{broken}}(m) + N_{\text{formed}}(m) + N_{\text{order changed}}(m). \tag{3}$$

This shared score differs from optional metal-specific thresholds elsewhere in the software. SLAP H assignments are refined within saved label pools. GRAFT minima concern saved representatives, not a certified optimum over all compressed families.

Heavy classes are compared using exact graph certificates with score-preserving endpoint equivalences. Missing representatives are queried against compressed families when the budget permits; unresolved membership stays unknown. The main comparison uses tolerance 1.0, ten orderings, and cap 100 throughout. A previous case-123 cap-200 follow-up remains separate.

2.8 Compute accounting and frozen baseline

The seed ablation uses frozen native-reuse engine `98b01b1`, with eight CPUs per directional GRAFT call. The timing subset contains 1,807 Golden reactions completed by every configuration. Instrumented parent-plus-child GRAFT CPU excludes measured saving/loading but includes process and communication setup. SLAP's workflow timer excludes worker startup. Interrupted work without final readings is excluded from this paired successful-work comparison. These are scoped compute measurements, not total resources or equal-resource latency. Recovery always uses 1,851 records.

Earlier engine `bb0a7d7` provides a separate frozen recovery and concrete-candidate collection experiment. Its outputs are not unioned into the seed ablation. Saved artifacts include inputs, source hashes, cut archives, statuses, witnesses, and timings. Figures are generated from machine-readable report snapshots.

3 Results

3.1 Continuous growth retains and resolves competing placements

An uncut replay makes the evolving candidate set explicit on 18-atom carbocation case 64 (Figure 2). The first ordering starts at source atom 13 with five compressed candidates. Growth increases the count to 42 before additional context reduces it to two distinct saturated placements. The same fragment keeps extending as its placement set expands and contracts; saturation then supplies the placements on which subsequent search can branch. Candidate counts measure compressed states, not atom permutations.

Movie 1 and the interactive viewer replay this trajectory and expose its two terminal representatives. Movie 2 follows TEMPO case 1 through eight deferred-boundary observations and four fragment decisions. Both use one ordering, no sweep, tolerance 1.0, and cap 100 with the event-enabled Python path of the frozen source. These illustrative runs do not contribute to benchmark totals or timing. Animation time indexes algorithm events, not physical reaction time.

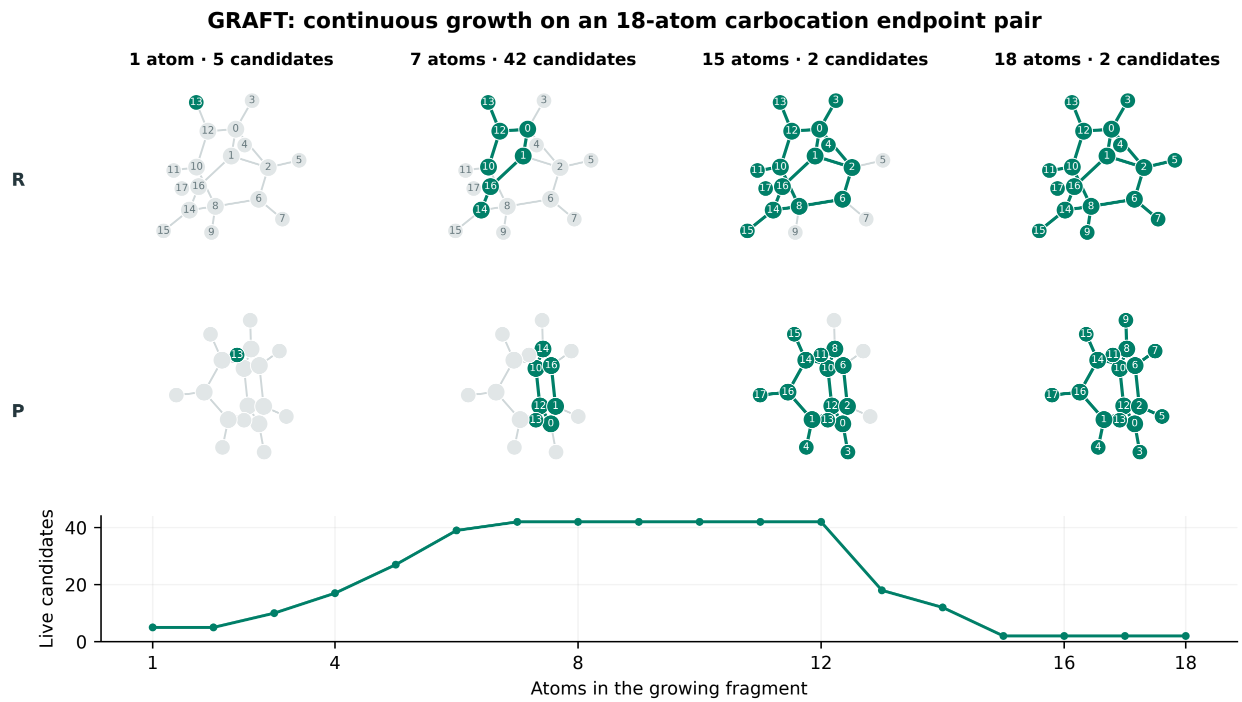

Figure 2 An actual growth trajectory. Source (R) and target (P) snapshots display a representative at selected stages of case 64. Green marks the current fragment and its image; assigned target labels report source identity. Below: live compressed candidates versus atoms added. Representative choices can change between steps. Coordinates are projected for illustration; two placements survive saturation.

Table 1 Seed ablation under the frozen native-reuse engine. Outcomes use all 1,851 Golden records. CPU sums both directions and all cuts on the same 1,807 completed reactions.

Configuration	Recovered	%	Absent	Unknown	CPU s/reaction
1 seed order	1,833	99.03	17	1	11.72
2 seed orders	1,834	99.08	14	3	20.85
3 seed orders	1,836	99.19	10	5	29.77
10 seed orders	1,836	99.19	5	10	87.87

3.2 Golden reference recovery with few seed orderings

Under the frozen benchmark baseline, GRAFT with one ordering per cut recovers 1,833 references (99.03%), with 17 absences and one unknown. Two recover 1,834 (99.08%), and three recover 1,836 (99.19%). The ten-order baseline also verifies 1,836, although case sets and unknown outcomes differ (Table 1). Each fresh one-, two-, and three-order experiment completes all 3,982 directional searches and 3,982 analysis processes across both collections, without a hard watchdog.

Mean paired costs are 11.72, 20.85, 29.77, and 87.87 CPU-seconds per reaction for one, two, three, and ten orderings. One ordering costs 7.50-fold less than ten, retaining verified recovery within three records of the ten-order total (Figure 3).

Equal totals do not imply identical coverage. Relative to ten orders, one order no longer verifies seven cases, including one now unknown, but verifies four previously unknown cases. Those gains reflect completed search or verification, not a superset explored by fewer orderings. No cross-setting union contributes to reported recovery.

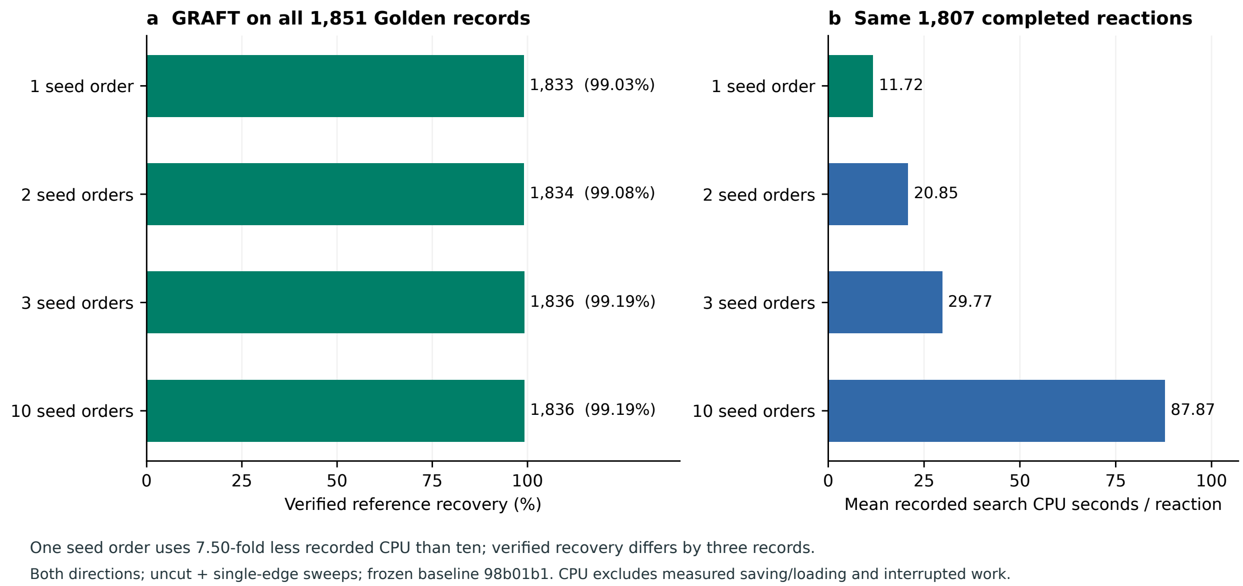

Figure 3 GRAFT reference recovery and the cost of seed diversity. (a) Verified reference witnesses among retained alternatives on all 1,851 Golden records. (b) Mean recorded search CPU for the same 1,807 reactions completed by all compared settings. All settings use both directions, the uncut graph and single-edge sweeps, tolerance 1.0, and cap 100 under frozen engine `98b01b1`. CPU includes process setup and excludes measured saving/loading and interrupted work.

3.3 Alternative mappings from native XYZ/WBO inputs

All four seed settings return full explicit-atom mappings for all 140 XYZ/WBO cases. Best saved scores for one, two, and three orders agree with ten in 139 cases. In case 123 the counts are respectively 27, 25, 25, and 19. Mapping availability is stable under seed reduction, whereas representative score can change.

In the separate tolerance-1.0 comparison with native XYZ SLAP, at ten orders and strict cap 100, best scores tie in 133 cases; GRAFT is lower in six and SLAP in one (Figure 4). The separate cap-200 follow-up is reported in the supporting information; all main comparisons retain cap 100.

Of 176 SLAP heavy classes, 159 are represented in strict-cap GRAFT families, 12 are excluded from those archives, and five are unresolved. Conversely, 97 cases have an extra GRAFT heavy pattern absent from saved SLAP at the lowest observed GRAFT score. This rises to 99 within one additional event, 132 within two, and 138 within five. Score agreement does not imply equal families. Neither method is shown to contain the other's complete output space; missing annotations prevent a chemical-accuracy claim.

3.4 Supporting ablation: single-edge constraint relaxation

With both directions and bond modes fixed, SLAP recovers 1,661/1,851 references (89.74%) in the uncut control and 1,795 (96.97%) after single-edge sweeping: 134 additional cases, or 7.24 percentage points (Figure S1). The 56 nonrecoveries comprise 52 completed misses and four watchdog-limited unresolved cases; all remain in the denominator. The gain supports individual edge deletions as a useful source of alternative search trajectories. Recovery is the union over cuts, rather than the output of one privileged cut.

This is a supporting ablation of the cut policy in SLAP. Its cuts perturb graph initialization and the assignment objective, whereas GRAFT's cuts relax growth constraints; it does not isolate GRAFT's branching or deduplication contributions. In the main frozen-baseline experiment, one-order GRAFT recovers 1,833 references (99.03%), 38 more than the SLAP sweep. Recorded mean CPU on the common subset is 11.72 seconds for GRAFT and 17.62 for SLAP, subject to the accounting differences specified in Methods. Detailed sweep completion and the earlier expanded-SLAP result are reported separately in the supporting information.

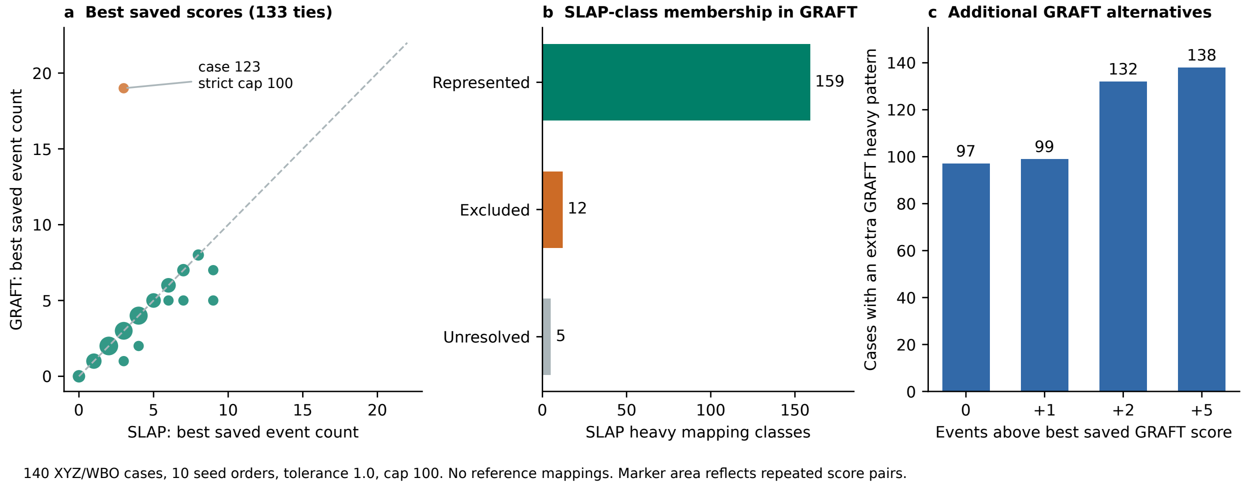

Figure 4 Saved outputs on 140 XYZ/WBO cases. All panels use tolerance 1.0, ten orders, and strict cap 100. (a) Best representative scores under a shared full-H WBO score; the dashed line denotes equality. (b) Membership of 176 SLAP heavy classes in saved GRAFT families. Exclusion concerns these archives; unknowns are separate. (c) Cases with an extra GRAFT heavy pattern absent from saved SLAP within each event window. No reference mappings exist for this collection.

4 Discussion

GRAFT makes the evolution of a fragment's candidate placements central to atom mapping. Continued growth incorporates additional structural context before saturation, and surviving placements define conditional branches for the remaining atoms. Local candidate canonicalization and compatible state joins reduce repeated representations while preserving distinctions that may affect continuation. Completed families retain correlated symmetry and the decisions supporting each correspondence. Together, these operations provide an inspectable account of the alternatives actually found; exact deduplication does not make bounded search exhaustive.

The frozen-baseline results connect this representation to useful reference recovery and alternative retention. One seed ordering per cut verifies 99.03% of Golden references, compared with 99.19% for three or ten orderings. Its lower recorded CPU makes it a useful operating point for further validation. Equal recovery totals can conceal different case sets and verification unknowns, and the coordinate results show that representative score can change even when full mapping availability is unchanged. Applications requiring alternatives may therefore value coverage beyond the reference-witness metric.

The SLAP ablation supports single-edge relaxation as a practical search choice: a bounded perturbation per run can improve recovery when outputs are combined across cuts. This observation motivates the cut policy used with GRAFT while leaving the algorithmic contribution in continuous growth, conditional branching, and context-preserving deduplication. Matched component ablations would be needed to quantify the separate performance contribution of each operation.

Several limitations define scope. Golden is a finite curated set; a reference may specify only one plausible transformation, and results depend on equivalence criteria. The coordinate collection lacks annotated mappings and was inspected during development, so it is not an untouched external accuracy test. Finite schedules, greedy growth, support budgets, caps, and watchdogs limit exploration. Symmetry exactness concerns the configured graph relation rather than every physical notion of equivalence. Recorded CPU does not establish total campaign cost or hardware-normalized latency.

Low-event alternatives need chemical assessment. Event minimization provides neither an energy model nor mechanistic validation. Downstream alignment and TS interfaces offer routes to evaluate hypotheses, but their existence does not establish improved TS prediction. Expert annotation, complete-pipeline resource measurements, and prospective testing of a frozen operating point would strengthen that connection.

5 Conclusion

GRAFT is an atom-mapping algorithm built around continuous fragment growth, branching on saturated placements, and context-preserving deduplication. It retains structured mapping families with correlated symmetry and explicit search histories. The frozen benchmark baseline recovers 99.03% of Golden references with one seed ordering per cut and 99.19% with three, while native coordinate inputs demonstrate full mapping availability and additional low-event alternatives. A supporting single-edge ablation in SLAP strengthens the empirical case for bounded constraint relaxation. GRAFT's retained families provide mapping hypotheses that can be inspected and evaluated in subsequent mechanistic studies.

Acknowledgments

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Data and code availability

The implementation, frozen configurations, per-case evaluations, and compressed search archives are retained with this study. Figure data, source hashes, and scripts accompany this draft. Public repository and permanent archive identifiers will be specified before submission. Golden data are available from the original authors' repository (3).

Author contributions and competing interests

Varinia Bernales and Alán Aspuru-Guzik are the principal investigators. Individual contributions and competing-interest declarations will be completed and approved by all authors before submission.

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A Supporting Information

A.1 SLAP single-edge ablation and completion

Binary single-edge sweeps recover 1,785 Golden references and weighted sweeps recover 1,762; their complementary outputs yield 1,795 recoveries in union (Figure S1). SLAP's initialization is rebuilt after every cut, following the protocol in Methods.

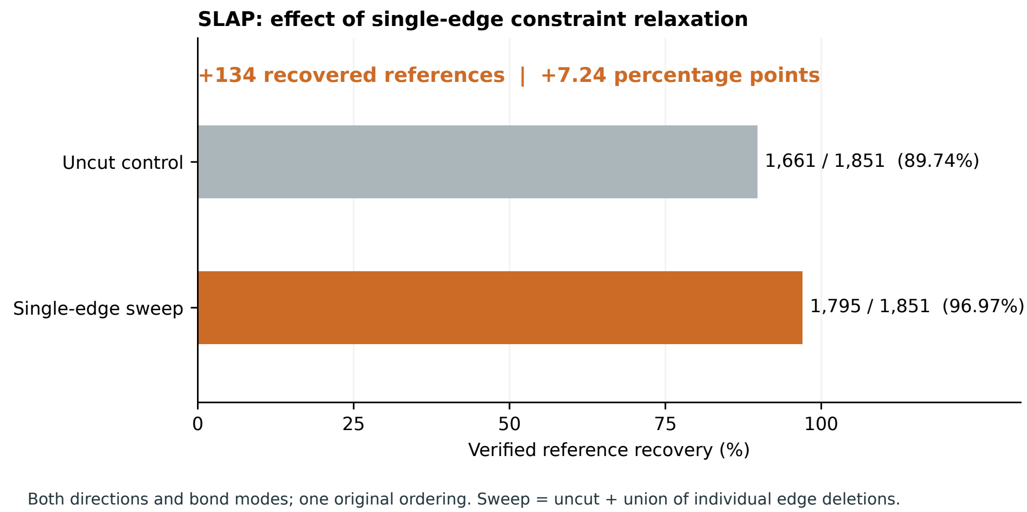

Figure S1 Supporting single-edge ablation in SLAP. The control and sweep use the same original ordering, both directions, and binary and weighted modes. The sweep unions the uncut output with outputs from individual source-edge deletions. Recovery increases by 134 cases (7.24 percentage points). All 1,851 records remain in the denominator, including incomplete variants. This experiment evaluates the cut policy in SLAP, not the individual components of GRAFT.

Of 355,104 planned calls, 352,210 are recorded: 352,208 succeeded and two had upstream exceptions. All returned mappings pass strict format and endpoint-preservation checks. Of 7,404 direction/mode variants, 65 reached the watchdog; their completed cuts remain included. Standalone nonrecoveries comprise 52 fully swept misses and four unresolved cases (1503, 1665, 1723, 1776), containing nine unfinished variants. Case 1665 timed out in both directions. Incomplete nonrecovery does not establish absence.

Earlier expanded SLAP uses an additional ordering budget and recovers 1,691 (91.36%). Its union with the sweep adds cases 1071 and 1595, reaching 1,797 (97.08%); 50 completed misses and four unresolved cases remain. This separate union is excluded from the ablation figure.

A.2 Default competitor reproduction

All 1,851 Golden records were submitted to the released configurations in Table 2. These are saved common-evaluator results, not literature numbers. SLAP's first output is not asserted to be a separately ranked top-1. Chython denotes its released ONNX implementation rather than a historical GraphormerMapper checkpoint.

Indigo returned 503 predictions with duplicate heavy-atom labels and seven with changed endpoint chemistry, in addition to exceptions. Duplicate labels were observed in its native API on an inspected example. The score should not be generalized to historical Indigo benchmarks. Chython returned 29 endpoint-modifying predictions. RDT required worker restart and retry after a JVM failure and broken-pipe cascade; 232 affected initial attempts remain archived. Three final RDT candidates failed validation.

A.3 Separate experimental versions

The seed ablation uses `98b01b1`; the earlier publication run uses `bb0a7d7`. The former's ten-order evaluation has 1,836 recoveries, five absences, and ten unknowns. The latter has 1,839 recoveries, nine absences, and

Table 2 Released implementations under the common evaluator. Correctness columns give counts out of 1,851. Output cardinalities and search budgets differ.

Implementation	First correct	Any correct	Exceptions
RXNMapper 0.4.3	1,547	1,547	0
LocalMapper 0.1.5	1,605	1,605	0
Chython 2.18 / chython-rxnmap 2.0	1,561	1,561	5
SLAPMapper 1.0.0, binary	1,399	1,591	0
SLAPMapper 1.0.0, weighted	1,401	1,564	0
Indigo 1.46.0	692	692	133
RDT 4.0.0	1,030	1,030	0 after retry

three unknowns. Similar high-level settings do not establish output equivalence. The earlier run completed 3,689 of 3,702 directional searches; 13 reached the watchdog. Its representative top-5 recovery is 1,789/1,851 (96.65%), distinct from top-5 family and event-window recovery.

Relative to ten orders, one order no longer verifies cases 44, 602, 850, 912, 1314, 1384, and 1568; 912 is unknown and the other six are absent. Two orders no longer verify 44, 602, 850, 1314, 1384, and 1568; 850 and 1568 are unknown. Both fresh runs verify 1033, 1786, 1799, and 1823, previously unknown. Identifiers are zero-based.

A.4 Concrete recovery beyond minimum event score: earlier engine

The earlier publication engine verifies 1,839/1,851 references (99.35%) bidirectionally, with nine absences and three unknowns. Its representative top-1 result is 1,434/1,851 (77.47%). This gap illustrates why family recovery cannot be presented as single-answer accuracy.

Extraction from those archives recovers all 1,839 as concrete candidates within six additional bond events of the best collected candidate (Figure S2). At minimum score alone, recovery is 1,692 (91.41%); allowing two additional events gives 1,817 (98.16%). This is an event window, not top-k. Extraction remains bounded: 20,916 visited families are unresolved and only 409 bidirectional records have exhaustive extraction certificates. The result shows useful higher-event candidates without establishing that all are chemically plausible.

A.5 Adaptive no-sweep search: a separate policy

This experiment uses revision `0ce0f17` and one CPU per direction. A balanced seed agenda uses saved budgets of 400, 1,600, and 6,400 growth operations and a 30 CPU-second soft limit per direction. It retains the native growth cap of 100 but replaces the synchronized frontier cap with an agenda budget. Its last saved checkpoint defines output. Budget completion does not imply agenda exhaustion or successful symbolic verification.

The separate adaptive no-sweep policy verifies 1,723/1,851 Golden references (93.08%) bidirectionally, with 33 verified absences and 95 unknowns. All 3,702 directional Golden searches complete under their budgets. On the 140 coordinate pairs, it returns full mappings for every case and matches the original ten-order sweep's best saved score in 138 cases; two scores are worse. Thus high mapping availability and score agreement on coordinate inputs do not establish equivalent reference recovery or retained families. This experiment does not isolate the effect of removing cuts from the original scheduler.

A.6 Accounting scopes

Across all Golden records, fresh one-, two-, and three-order searches consume 27,084.86, 49,161.91, and 70,570.73 instrumented CPU-seconds. Holdout totals are 1,475.45, 2,417.94, and 3,348.29 seconds. Full-workload totals differ from the common subset.

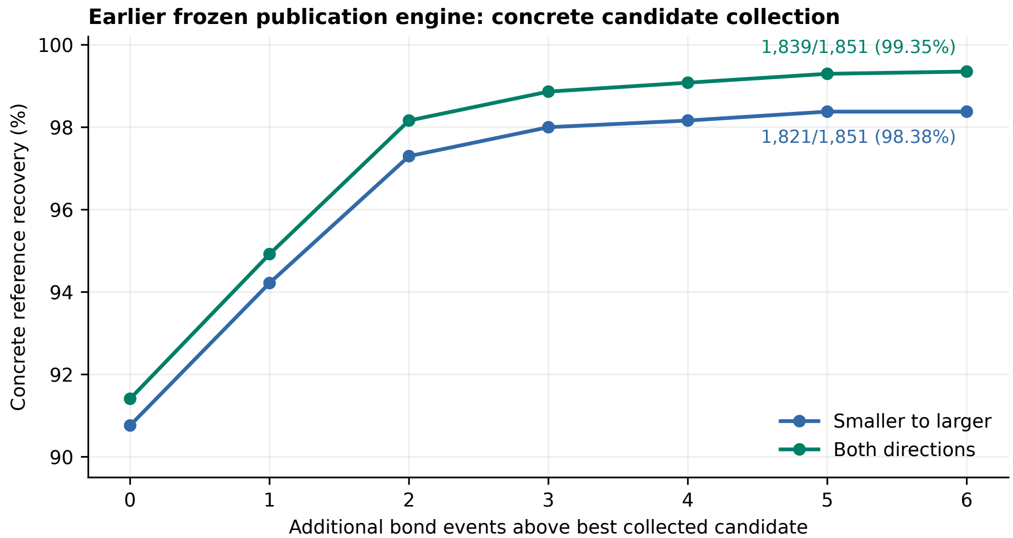

Figure S2 Recovery among collected concrete candidates. This uses the earlier publication engine, separately from the seed ablation. Event windows are relative to the best collected score at maximum coverage. All 1,851 records remain in the denominator. Extraction does not enumerate every represented mapping.

Completed SLAP sweep calls consume 14.479 CPU-hours of native mapping; mapping plus preparation and export uses 14.934 CPU-hours. Interrupted calls lack final readings. Scheduler TotalCPU fields cannot recover missing consumption on this cluster. Neither these totals nor reserved CPU-hours establish actual total campaign CPU. Mapping spans 8 minutes 9 seconds and separate evaluation spans 35 seconds under allocations different from GRAFT's. These spans are not a matched-latency comparison.

A.7 Holdout follow-ups

At tolerance 1.0 and cap 100, case 123 has no full R-to-P output and a 19-event P-to-R representative. Its separate cap-200 follow-up returns a three-event mapping, yielding 134 score ties, six GRAFT-lower cases, and no SLAP-lower cases. Substitution changes SLAP-class membership from 159 represented, 12 excluded, and five unknown to 160 represented, 11 excluded, and five unknown. It is excluded from the fixed-cap panels.

Earlier tolerance-0.5 and proposal-guided studies use distinct policies. In an exploratory uncut holdout ablation, one random ordering matches the full-search best score in 133/140 cases and ten orderings in 138/140. These are score-agreement observations, not Golden accuracy or proof of family equivalence.

A.8 Supplementary animations

Movie 1: continuous growth and retained placements. Carbocation example `pr16.carbocation_ts5` (case 64) displays actual extension events and two terminal representatives from an uncut, one-order replay. Changing assignments reflect candidate refinement, not physical reaction motion.

Movie 2: deferred boundaries and successive fragments. TEMPO example `pr1.tempo_ts10` (case 1) follows first-branch events through four fragment decisions and eight deferred-edge observations. A deferred boundary is search evidence, not automatically a chemically broken bond.

The standalone HTML viewer provides playback, event scrubbing, sampled witnesses, terminal selection, H display, and coordinate rotation. Extension views sample at most five witnesses, without enumerating all compressed assignments. Inputs, events, frozen source hashes, and the bookkeeping-binary hash accompany the figure data.